BEpiC: Binary Episodes for Meta-Learning Towards Better Generalization

1st Atik Faysal
Electrical and Computer Engineering
Rowan University
Glassboro, NJ
faysal24@rowan.edu

2nd Mohammad Rostami
Electrical and Computer Engineering
Rowan University
Glassboro, NJ
rostami23@students.rowan.edu

3rd Huaxia Wang
Electrical and Computer Engineering
Rowan University
Glassboro, NJ
wanghu@rowan.edu

4th Avimanyu Sahoo
Electrical and Computer Engineering
University of Alabama in Huntsville
Huntsville,AL
avimanyu.sahoo@uah.edu

5th Ryan Antle
R&D head
Baker Hughes
Oklahoma City, OK
ryan.antle@bakerhughes.com

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***Abstract*—In practical machine learning applications, image classification often involves discerning the category to which an object belongs. In this paper, we present a novel approach called Binary Episode Classifier (BEpiC) within the context of meta-learning. BEpiC aims to optimize episode generation by strategically selecting training samples. The methodology involves training the initial class with a comprehensive set of similar images while the contrasting class is exposed to a diverse array of dissimilar images. We create highly dissimilar image sets by randomly forming multiple image clusters. Identifying the cluster centers and selecting the representative image closest to each center facilitates the determination of the among clusters. The distance of these images from the other class is then calculated. The cluster whose representative image exhibits the greatest distance is chosen as the definitive class. Our proposed method showcases significant efficacy for binary meta-learning classification across diverse classes. While our experimentation primarily focused on Model-Agnostic Meta-Learning (MAML), the adaptability of the episode generation strategy extends to a spectrum of meta-learning classifiers. Empirical findings substantiate that the proposed method attains remarkable accuracy in one-shot classification scenarios and moderately higher accuracy in few-shot classification tasks.**

***Index Terms*—meta-learning, few-shot learning, image classification, supervised learning**

I. Introduction

A fundamental task in image classification problems is to determine with high confidence whether an object belongs to a specific class. These binary classifiers have applications in several safety-critical systems, such as self-driving cars for obstacle detection and automated visual inspection of aircraft engines for defects. For instance, autonomous vehicles must make split-second navigational decisions upon detecting an obstacle irrespective of the type of obstacle, such as a pedestrian, bike, or any other object.

On the other hand, training a binary classifier, using datasets with limited categories poses a significant hurdle. This challenge becomes apparent during testing when confronted with unfamiliar images. On the other hand, real-world scenarios often present an infinite array of new classes. Similarly, traditional supervised learning methods for multiclass classification fall short in classifying new classes without additional retraining [1].

Meta-learning and few-shot learning [2] offer promising solutions to address the challenges posed by new or unseen classes in classification problems by learning from a few samples. Most meta-learnings rely on episodic-based learning where an episode is defined as each training run involving different classes. In episodes, the training data is divided into support and query sets. The support sets train the meta-trainer classifier with a few samples for each task. The query sets, which consist of new unseen examples, evaluate the classifier's performance post-training for optimizing the meta-trainer parameters. This dynamic adjustment enhances the adaptability and generalization capabilities of the model [3], enabling it to handle new classes with limited examples effectively.

MAML is a meta-learning approach that aims to discover optimal initial parameters across task embeddings for rapid adaptation [4]. Several variants of MAML, such as Reptile [5], Meta-SGD [6], and LEO [7] are also developed recently with the common goal of achieving efficient adaptation. On the other hand, metric-based meta-learning approaches, such as prototypical [8] and relation networks [9], learn a metric space where classification is executed through distance computations and similarity, respectively, to prototype representations of each class. While meta-learning has effectively tackled the challenge of classifying new classes with minimal data, to the best of our knowledge, no studies have specifically addressed binary classification scenarios where one class encompasses multiple categories. The Regular meta-learning episodes only contain samples of the same distribution in that particular class. Therefore, special attention needs to be paid to making it more adaptable for our specific application.

Motivated by this critical, unique, and unexplored dimension in meta-learning, in this paper, we introduce a Binary Episode Classifier (BEpiC) in the meta-learning framework. The methodology

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involves training the initial class with a comprehensive set of similar images while the contrasting class is exposed to a diverse array of dissimilar images. Unlike the conventional episode generation methods, referred to "Regular" episodes, BEpiC enriches variability in each training episode by selecting images with maximum class distance. Thus, BEpiC assists the meta-trainer through abundant training samples for improved selection. For binary classification, in the first class, we curate samples from the same category, while the second class is populated with samples from diverse sources distinctly dissimilar to those in the first class.

To formulate the BEpiC episodes, we randomly generated multiple data clusters containing samples from numerous distributions. We reduced sample dimensionality using Principal Component Analysis (PCA) [10] to identify each cluster's centroid. The sample nearest to the centroid was designated as the representative of its respective class. To quantify class dissimilarity, we employed the Oriented FAST and Rotated BRIEF (ORB) [11] to identify the most dissimilar representative image from the representative image of the other class, which contains only similar samples.

In addition, we integrate the "Regular" and "BEpiC" techniques, referred to as "Regular+BEpiC," to leverage the complete range of accessible data. We performed experimental studies for both episode generation techniques using Caltec256 and miniImageNet datasets. The results illustrate that the BEpiC and Regular+BEpiC techniques notably boost classifier accuracy and robustness, affirming their effectiveness in meta-learning accuracy. The ***main contributions*** of the paper are 1) a novel episode generation technique for the binary meta-learning approach to assist the meta-trainer for improved classification, 2) the generation of multiple class clusters randomly, 3) the reduction in sample dimensionality to identify each cluster's centroid, and 4) quantify class dissimilarity by using Oriented FAST and Rotated BRIEF (ORB) approach.

The remainder of the paper is organized as follows. Section II presents the background on MAML and the problem formulation. Section III provides a detailed discussion of the proposed methodology followed by experimental evaluation and results in Sections IV and V. The ablation study is presented in Section VI followed by conclusion in Section VII.

II. Background and Problem Statement

This section offers a theoretical background. We initiate our discussion with an overview of the MAML classifier, followed by a comprehensive problem description.

Meta-learning aims to refine a model's initial parameters, facilitating optimal performance on a new task. This is achieved through parameter adjustments via gradient steps computed with a limited dataset. In the case of MAML, this strategy is adopted to obtain a generalized performance. MAML exhibits effectiveness across regression, classification, and reinforcement tasks. Given its superior performance, we have chosen MAML as the primary and only classifier for our work.

MAML is a meta-learning approach that is optimization-based and tries to find the initialization parameters, Θ, for fast adaptation. For a particular task $\tau_i = (D^{train}, D^{val})$, where D^{train} and D^{val} demote the train and validation support and query sets [12], MAML computes the updated parameters $\Theta_i^{'}$ by applying gradient descent, given by

$$\Theta_i^{'} = \Theta - \alpha \nabla_{\Theta} \mathcal{L}_{\tau_i}^{train}(f_{\Theta}) \tag{1}$$

where α is the learning rate of the inner loop and $\mathcal{L}_{\tau_i}^{train}$ is the training loss in the inner loop. ∇_{Θ} is the gradient operator with respect to the model parameters Θ.

The initial parameters are used to do forward propagation on the validation set and update the meta-learning parameters via a second-order gradient descent, given by

$$\Theta \leftarrow \Theta - \eta \nabla_{\Theta} \sum_{\tau_i \sim p(\tau)} \mathcal{L}_{\tau_i}^{val}(f_{\Theta_i^{'}}) \tag{2}$$

where η is the learning rate and $\mathcal{L}_{\tau_i}^{val}$ is the validation loss in the outer loop.

In meta-learning, conventional episode generation involves randomly selecting classes from the dataset, followed by sampling instances from those chosen classes. The inter-class or intra-class distance is never considered when generating the episodes. However, it is crucial in binary meta-classification, where a variety of samples may need to be classified against one class. On the other hand, formulating such episodes with dissimilarity is challenging since there are so many possible combinations, and calculating the distance between each sample is computationally expensive.

Therefore, the problem at hand is to generate episodes that can use the intra- or inter-class dissimilar to classify a variety of classes against one class. A solution to this problem is presented in the next section by introducing BEpiC classifier. It is important to note that our proposed approach of selecting class representatives cut down the computational expenses to a great extent.

III. Proposed Methodology

In this section, we propose a solution to the problem of episode generation along with the development of the BEpiC algorithm. The solution involves formulating the classes with the best samples selected using PCA and ORB. A visual representation of our episode generation process is illustrated in Figure 1. Our implementation can be found at https://github.com/atik666/BEpiC.

A. Samples Selection

Since the goal of the episode generation is to diversify the samples, sample selection identifies samples that maximize the inter-class distance, which is one of the contributions of our paper. However, computing the distance between every sample across different classes poses a substantial computational challenge, especially for large datasets. To address this, we employ a strategy of randomly designing multiple episodes. For instance, considering a class (referred to as the "cat" class for clarity), we determine which mixture of other classes would yield the most dissimilar set.

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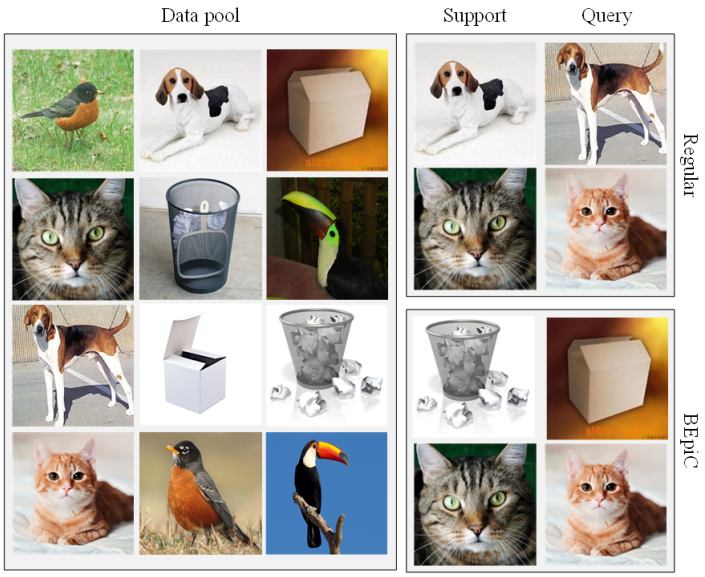


Fig. 1. Episode generation of Regular and BEpiC from the data pool. In BEpiC, we select the most dissimilar samples.

The sample selection process involves calculating the centroid of each episode cluster. It is achieved through PCA for dimensionality reduction and mean calculation to determine the centroid. The Euclidean distance metric is then applied to identify the sample closest to the centroid, which is subsequently designated as the class representative. To streamline computations, we later only compare the most centroid images with the ones from the "cat" class. This strategic reduction in the number of comparisons contributes to managing the computational complexity of analyzing large datasets.

We employ the ORB feature-matching algorithm to assess the dissimilarity between samples from the two classes. We select the one with the highest dissimilarity score from a pool of numerous samples. In cases where multiple outputs exhibit the same lowest dissimilarity, we prioritize the first one on the list. This approach ensures we pinpoint the most dissimilar pair of samples, effectively identifying representative class instances. In the following subsection, we provide a brief description of PCA and ORB.

1) PCA: Given a dataset $X \in \mathbb{R}^{n \times p}$ with n observations and p variables, PCA seeks to find a set of orthogonal vectors, that capture the most variance in the data. The first step in PCA is to subtract the mean of each variable from the dataset X, resulting in,

$$X_{\text{centered}} = X - \bar{X} \tag{3}$$

Next, we compute the covariance matrix using the center,

$$C = \frac{1}{n-1} X_{\text{centered}}^T X_{\text{centered}} \tag{4}$$

We solve $CV = V\Lambda$ to get eigenvectors V and eigenvalues Λ and project the data onto the new coordinate system $Y = X_{\text{centered}}V$. We select the top k eigenvectors to reduce the dataset dimensionality: $Y_k = X_{\text{centered}}V_k$.

The variance explained by the i-th principal component is $\frac{\lambda_i}{\sum_{j=1}^{p} \lambda_j}$, where λ_i is the i-th eigenvalue.

2) ORB: ORB calculates the angle intensity based on its center offset, making it suitable for determining orientation., given by

$$\mathrm{m}_{pq} = \sum_{x,y} x^p y^q I(x,y) \tag{5}$$

where m is referred to as the moment patch, p and q are the sequences of moments comparable to mechanical moments, and (x, y) is the pixel coordinate keypoint of image, I. The centroid can then be determined using the moment in (5) as

$$C = \left(\frac{m_{10}}{m_{00}}, \frac{m_{01}}{m_{00}} \right) \tag{6}$$

The centroid will be located using the image's pixel matrix, designated as $\{m_{00}, m_{01}, \cdots\}$. The vector is created by connecting the centroid C with the angle's center. In other words, the patch's orientation can be computed as

$$\Theta_{orient} = arctan(m_{01}, m_{10}) \tag{7}$$

where Θ_{orient} is the orientation angle based on the gradients. These selected samples craft the BEpiC episodes using any dataset that contains numerous classes.

B. Dataset

We conduct experiments on two popular datasets in meta-learning, Caltech 256 [13] and miniImageNet [14]. The statistics for the two datasets can be found in Table I.

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TABLE I
STATISTICS FROM THE TWO IMAGE DATASETS.

Dataset	#Classes	#Training	#Validation	#Test	#Total samples	#Samples/Class
Caltech 256	256	150	56	50	30,607	at least 80
miniImageNet	100	64	16	20	60,000	600

1) Caltech 256: The Caltech 256 dataset is considered an enhancement over its predecessor, the Caltech 101 dataset, due to features such as larger category sizes and the inclusion of new and more extensive clutter categories [15]. This dataset is ideal for developing visual recognition and classification models, comprising 30,607 real-world images across 257 classes of varying sizes (256 object classes plus an additional clutter class). Each class includes 100 training images and 20 test images.

2) miniImageNet: miniImagenet, a subset derived from the extensive ImageNet dataset, is specifically tailored for few-shot learning tasks. The unique challenge posed by miniImagenet lies in its demand for machine learning models to generalize effectively with only a limited number of training examples. This dataset consists of 100 classes, each accompanied by 600 images. We allocate 64 classes for training, 16 classes for validation, and 20 classes for testing. This division ensures a balanced distribution of classes across different sets, enabling a comprehensive assessment of model performance.

The experimental evaluation and results are presented in the next sections, which employ the sampling process and dataset.

IV. Experimental Evaluation

We conducted a comprehensive evaluation of our model, focusing on its test accuracy to ensure a thorough examination of its performance. We extended our analysis to encompass the accuracy of various model configurations. The evaluation employed terminologies commonly used in meta-learning research, specifically the n-way k-shot learning framework, where n-way denotes the number of classes and k-shot indicates the number of samples the classifier encounters in each training episode.

For binary classification, the number of ways is fixed as 2-way. However, we vary the k-shot classification to explore different configurations with $1, 5, 20$ values. Across all experiments, the total number of epochs is set at 10, each comprising $10,000$ training episodes and 100 test episodes.

For every input, the image size is standardized to $(84, 84, 3)$, ensuring uniformity in the dataset preprocessing. This consistent approach to experimental design allows for a thorough investigation of the model's performance under distinct k-shot scenarios, shedding light on its adaptability and effectiveness in binary classification tasks.

We experimented with two configurations: 1) exclusively utilizing the BEpiC setup and 2) incorporating both BEpiC and Regular setups for episode generation. Our findings indicate that the BEpiC configuration exhibits superior generalization compared to the sole use of regular task generation. Moreover, a significant performance enhancement is observed when combining BEpiC with the Regular setup, particularly with increased episodes and shots in each task.

Table II provides insights into the classifier's performance on the BEpiC test set, where the test class comprises samples from the most dissimilar classes. On the other hand, Table III outlines the classifier's performance on the BEpiC test set in conjunction with the Regular test data. Lastly, Table IV provides the classifier's performance on random test sets, where both the training and test sets were designed randomly without adhering to our proposed method, implying that the largest inter-class distance was not a focal consideration. These comparative tables underscore the effectiveness and advantages of our proposed approach.

V. Results and Discussion

The findings from Table II provide compelling evidence supporting the superiority of BEpiC over Regular episodes in 1-shot classification scenarios. In this context, where test samples encompass any two different categories from the real world, our proposed MAML training method consistently outperforms the conventional approach for both datasets. The subsequent amalgamation of Regular and BEpiC setups results in a noteworthy 2% and 1.5% improvement in accuracy for 1-shot classifications on both datasets. This enhancement can be attributed, in part, to the doubled sample size, indicating that the classifier operates more effectively on the intended test samples in this combined arrangement. Expanding the evaluation to 5-shot and 20-shot classifications reveals a consistent trend, with Regular+BEpiC achieving the highest accuracy in all cases for both Caltech 256 and miniImageNet datasets.

The insights from Table III introduce intriguing nuances to our analysis. In the context of Caltech 256, including Regular test samples alongside BEpiC samples led to a 0.5% improvement in accuracy for the Regular classifier in 1-shot classification. This improvement is intuitively understandable, as adding Regular test samples facilitated the Regular setup's classification task. Surprisingly, the MAML classifier tested on BEpiC and Regular test samples achieved 1% higher accuracy than the previous BEpiC-only output. This seemingly counterintuitive result is clarified by recognizing the reduced challenge in classifying Regular test samples due to minimal intra-class distance. Consequently, the classifier excelled in handling Regular samples alongside BEpiC samples, resulting in higher accuracy. Notably, the 20-shot Regular Caltech 256 classification achieved the highest accuracy in this setup, suggesting that Regular test samples favored the

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TABLE II
PERFORMANCE OF THE CLASSIFIER ON THE BEPIC TEST DATASET.

Train methods	Test accuracy (%)					
	Caltech 256			miniImageNet		
	1-shot	5-shot	20-shot	1-shot	5-shot	20-shot
Regular	60.50	70.90	81.05	63.50	70.80	75.25
BEpiC	**61.00**	**71.90**	**81.50**	**64.00**	**71.00**	**76.66**
Regular+BEpiC	**63.50**	**72.80**	**81.80**	**65.00**	**71.40**	**76.66**

TABLE III
PERFORMANCE OF THE CLASSIFIER ON THE REGULAR+BEPIC TEST DATASET.

Train methods	Test accuracy (%)					
	Caltech 256			miniImageNet		
	1-shot	5-shot	20-shot	1-shot	5-shot	20-shot
Regular	61.00	70.00	**81.50**	65.50	69.60	76.46
BEpiC	**62.00**	**70.40**	80.96	65.00	**70.50**	**76.70**
Regular+BEpiC	**63.00**	**71.97**	81.25	**67.00**	**71.00**	**76.83**

TABLE IV
PERFORMANCE OF THE CLASSIFIER ON THE RANDOM TEST DATASET.

Train methods	Test accuracy (%)					
	Caltech 256			miniImageNet		
	1-shot	5-shot	20-shot	1-shot	5-shot	20-shot
Regular	57.50	70.70	80.70	62.40	**70.20**	74.83
BEpiC (random)	**58.00**	**71.00**	**81.25**	**63.70**	70.10	**75.33**
Regular+BEpiC (random)	**61.00**	**71.30**	**81.50**	**64.90**	**71.16**	**76.23**

Regular approach in achieving higher accuracy in 20-shot scenarios. However, for 5-shot learning, a drop in accuracy across all classifiers indicates the proposed method's increased suitability when there is greater variety in test samples. For miniImageNet, Regular+BEpiC consistently achieved the highest accuracy, except in 1-shot scenarios where BEpiC achieved slightly lower accuracy, with Regular achieving a 0.5% advantage.

Table IV serves as a cautionary demonstration of the repercussions of disregarding the proposed method. The outputs reveal the necessity of our proposed approach by employing random samples for training episodes, irrespective of their inter-class distance, and testing with a BEpiC setup using random samples. Lower accuracy in 1-shot learning across all outputs compared to Table II reinforces the importance of strategically selecting samples with higher inter-class distance. This trend persists in 20-shot and 5-shot Caltech 256 scenarios. However, in 5-shot miniImageNet, Regular training outperforms BEpiC (random), affirming the suitability of our proposed method, mainly when training on BEpiC samples with significant intra-class distance.

In summary, the evidence presented consistently advocates for the effectiveness of the proposed BEpiC methodology. Regular+BEpiC emerges as a robust approach, showcasing superior performance across various few-shot learning scenarios compared to conventional and random training methods.

VI. ABLATION STUDY

In our investigation of classifier performance, we systematically varied the number of clusters for the negative class to discern its impact. The choice of cluster count plays a pivotal role

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in influencing the selection of the most distant samples while considering computational costs. Striking the right balance is crucial for optimizing our research outcomes.

Table V showcases the classifier's performance with the negative classes drawn from 5 clusters. The results, compared against those from BEpiC in Table II, are evaluated based on the change in test accuracy (%). Positive change indicates improved accuracy, while negative values signify a decline. For 5 clusters, there is a consistent drop in test accuracy across all shot learning scenarios, ranging from 0.1% to 0.66%. This implies a discernible reduction in classifier performance across various settings when employing 5 clusters.

In Table VI, we extend our analysis to 20 negative class clusters, observing a mixed performance outcome. While some scenarios show improvement, others exhibit marginal decreases. Factors such as multiple clusters being equidistant may contribute to the performance fluctuations. The change in accuracy spans from -0.66% to 0.27%. However, the absence of consistent accuracy improvement, coupled with a doubling of computational time, leads us to choose 10 clusters as the optimal value.

Our exploration underscores the delicate trade-off between the number of clusters, computational efficiency, and classifier performance. The choice of 10 clusters emerges as a judicious compromise, balancing enhancing accuracy and managing computational costs in our research endeavors.

TABLE V
PERFORMANCE OF THE CLASSIFIER ON THE RANDOM TEST DATASET.

5 Clusters	**Test accuracy (%)**					
	Caltech 256			**miniImageNet**		
	1-shot	**5-shot**	**20-shot**	**1-shot**	**5-shot**	**20-shot**
BEpiC	60.72	71.32	81.30	63.67	70.90	76.00
(%) change	-0.28	-0.58	-0.20	-0.33	-0.10	-0.66

TABLE VI
PERFORMANCE OF THE CLASSIFIER ON THE RANDOM TEST DATASET.

20 Clusters	**Test accuracy (%)**					
	Caltech 256			**miniImageNet**		
	1-shot	**5-shot**	**20-shot**	**1-shot**	**5-shot**	**20-shot**
BEpiC	61.16	71.83	81.67	64.11	71.27	76.00
(%) change	0.16	-0.07	0.17	0.11	0.27	-0.66

VII. CONCLUSION AND FUTURE WORK

This study introduces a novel episode generation strategy for binary classification tasks that distinguish between positive and negative classes and comprise various samples. The proposed approach leverages statistical analysis to curate training episodes, emphasizing samples with the highest inter-class distance. The results underscore the proposed method's robustness in effectively identifying test samples using BEpiC episodes. Across all scenarios, the proposed method consistently outperforms classifiers trained through conventional methods, showcasing superior accuracy.

An additional enhancement to the proposed methodology involves the integration of BEpiC with Regular training samples. This hybrid approach adds robustness to the classifiers and improves accuracy. The findings emphasize the potential of combining different training setups to achieve superior performance in binary classification tasks.

This study aims to extend the classifier's capabilities to handle multi-class classification scenarios as part of future work. Additionally, efforts will be directed toward contributing to the evolution of the classifier's architecture to mitigate any overfitting issues. These advancements are expected to broaden the applicability and effectiveness of the proposed meta-learning strategy, fostering its utility in diverse classification tasks.

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